

Class Meeting Handout #3

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Today we will strive to review and master the chemical ballet of electron movement. Are your arrows correct? Are your intermediates reasonable?

The Building Blocks of Mechanisms

All organic mechanisms can be built from simple elementary reactions (dissociation, addition, substitution, etc.) Today, we will review some simple reactions from your 2nd-year organic chemistry experience.¹

Review: Mechanism Practice – 40 minutes

Answer the following problems. Use structures with correct *curved arrows* and words to describe each reaction.

1. Write the mechanism for an S_N1 reaction. Choose a reaction where the S_N1 pathway predominates.
2. Write the mechanism for an S_N2 reaction. Again, choose an example where this pathway is predominant.
3. Write the mechanism for an E1 reaction. As above, pick a good example.
4. Write the mechanism for an E2 reaction. Present a representative reaction.
5. Write the mechanism for base-catalyzed hydrolysis of an ester.
6. Write the mechanism for acid-catalyzed hydrolysis of an ester.
7. Now write the mechanism for acid-catalyzed condensation of an alcohol and a carboxylic acid to make an ester. Is it similar to the above? Why is the base-catalyzed option for ester synthesis not a thing?
8. Write the mechanism for electrophilic aromatic substitution.
9. Write the mechanism for mutarotation of glucopyranose in water as shown in Figure 1. Give three versions: in acid, in base and in neutral conditions.
10. Write the mechanism for imine formation under acidic, basic and neutral conditions. Which mechanism is faster?

This document was produced using the L^AT_EX typesetting language with the Tufte-handout document class. Chemical diagrams were prepared in ChemDoodle,

¹ McMurray's *Organic Chemistry* would be a good place to look for review material.

You may have observed that the class today planned 40 minutes of activities, that leaves 10 minutes for questions about previous classes, textbook problems, the assignment, or life in general. I will try to keep time available for questions in almost all class meetings from now on. If there are no questions, I will add course material to fill the time. You can help minimize your workload by asking questions.

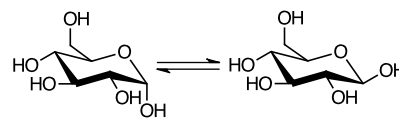


Figure 1: Mutarotation of glucose.↑

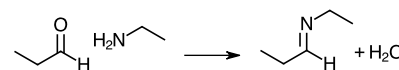


Figure 2: Condensation of an aldehyde and an amine to make an imine.↑

Learning Goals

There is more to a class meeting than what we do in 50 minutes of class time. After participating in the above exercises and completing the requirements described on the Moodle site, you will have achieved the following...

1. Able to use curved arrows to show the flow of electrons in an elementary reaction.
2. Able to propose a reasonable mechanism for a the muti-step reaction.
3. Able to evaluate a mechanism and estimate if a proposed intermediate is likely or unlikely to form.
4. Understand the connection between arrow pushing and transition state structure.

Looking Ahead

We will be using the skills that we reviewed today during the next class meeting to interpret reactions that you have seen used in organic synthesis examples presented in 2nd-year and 3rd-year organic chemistry courses. Understanding mechanism is essential in this course. Later, we will be introducing experimental techniques that can probe the unknowable structure of a transition state and support or disprove your arrow-pushing proposals. The first step in investigating a reaction is proposing a hypothesis for the mechanism.